Solution

Unofficial solution written by HaslaLab; all values were measured on the organisers' FBG-Writer model.

A.1. During one period $1/f$ of the pulses the fiber moves by v/f, and each pulse leaves one defect, so

$$\boxed{\Lambda = \frac{v}{f}}$$

A.2. From $\lambda_B = 2n_c\Lambda = 2n_c v/f$,

$$v_0 = \frac{f\lambda_B}{2n_c} = \frac{1.00\cdot 10^{3}\ \mathrm{s^{-1}}\cdot 1550\ \mathrm{nm}}{2\cdot 1.45} \approx 5.34\cdot 10^{-4}\ \mathrm{m/s}.$$

$$\boxed{v_0 \approx 0.5345\ \mathrm{mm/s}}$$

A.3. The grating written with $\Delta Y = 0.0\ \mu$m, $v = v_0 = 0.5345$ mm/s, $L = 50$ mm has its transmission minimum at

$$\boxed{\lambda_B = 1543.64\ \mathrm{nm}}$$

A.4. The period of this grating is $\Lambda = v_0/f = 0.5345\ \mathrm{mm/s}/1.00\ \mathrm{kHz} = 534.5$ nm, so

$$\boxed{n_c = \frac{\lambda_B}{2\Lambda} = \frac{\lambda_B f}{2v_0} = 1.4440}$$

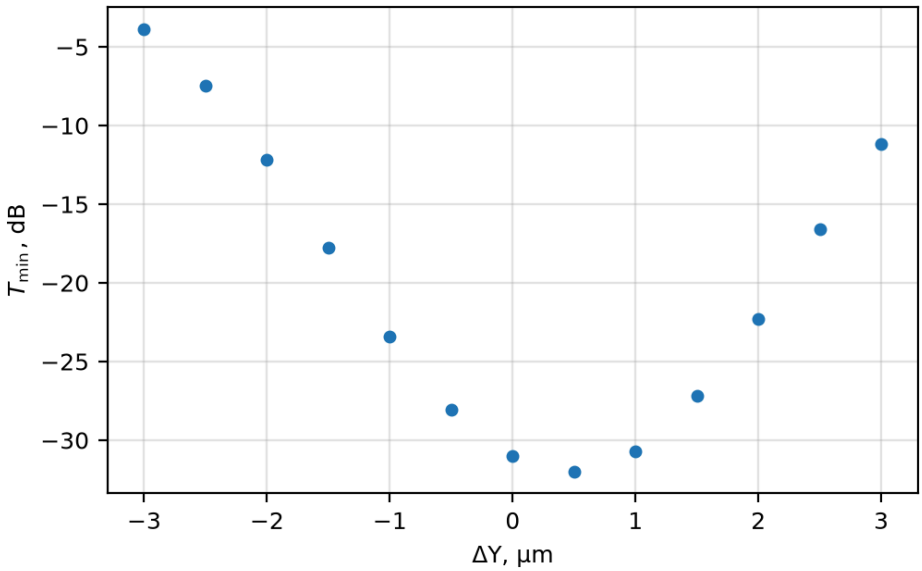
A.5.

$$\boxed{v_{1550} = \frac{f\cdot 1550\ \mathrm{nm}}{2n_c} = 0.5367\ \mathrm{mm/s}}$$

Writing a grating with this velocity indeed gives the resonance at 1550.00 nm.

A.6. Gratings written with $v = v_{1550}$, $L = 50$ mm. For each one the depth of the dip is read on a zoomed plot.

ΔY, µm	$T_{\min}$, dB	ΔY, µm	$T_{\min}$, dB
−3.00	−3.87	0.50	−31.98
−2.50	−7.48	1.00	−30.71
−2.00	−12.20	1.50	−27.16
−1.50	−17.77	2.00	−22.29
−1.00	−23.41	2.50	−16.57
−0.50	−28.04	3.00	−11.19
0.00	−31.01		

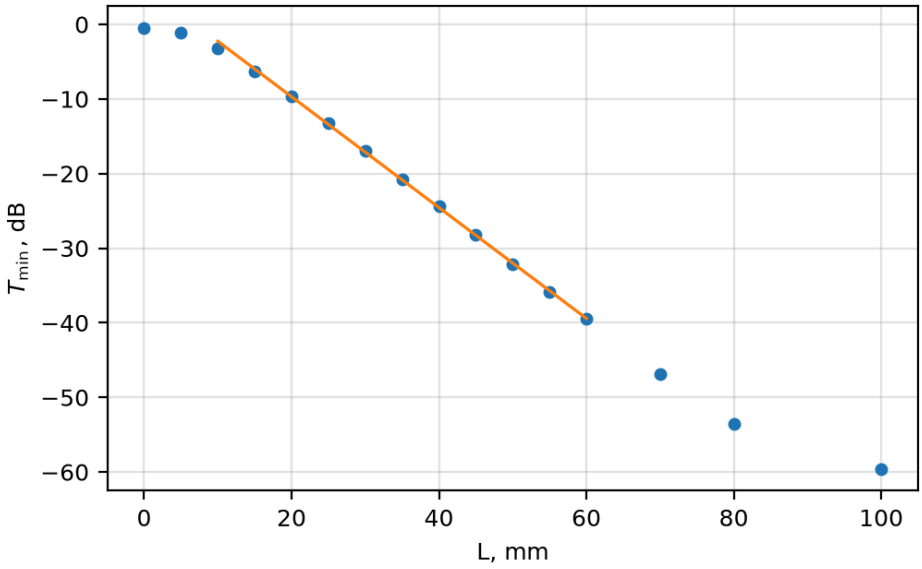


A.7. The dependence is symmetric about its minimum. A parabola through the three deepest points ($\Delta Y = 0.0,\ 0.5,\ 1.0\ \mu$m) has its vertex at

$$\boxed{\Delta Y_0 \approx 0.47\ \mu\text{m}}$$

A.8. Gratings written with $\Delta Y = \Delta Y_0$, $v = v_{1550}$ and different L.

L, mm	$T_{\min}$, dB	L, mm	$T_{\min}$, dB
0	−0.48	40	−24.40
5	−1.17	45	−28.23
10	−3.24	50	−32.10
15	−6.27	55	−35.86
20	−9.69	60	−39.48
25	−13.29	70	−46.90
30	−16.94	80	−53.55
35	−20.79	100	−59.56



For short gratings the dip is shallow and $T_{\min}$ is not yet linear in L; above $L \approx 70$ mm the dip reaches the background level (≈ -60 dB). Between 15 and 55 mm the dependence is linear:

$$\boxed{\frac{dT_{\min}}{dL} = -0.743\ \text{dB/mm}}$$

B.1. Differentiating the first equation and substituting the second,

$$a'' = i\varkappa b' = i\varkappa(-i\varkappa a) \quad \Rightarrow \quad \boxed{a''(z) = \varkappa^2 a(z)}$$

B.2. The general solution is $a = C_1\cosh\varkappa z + C_2\sinh\varkappa z$, and then $b = a'/(i\varkappa) = -i\,(C_1\sinh\varkappa z + C_2\cosh\varkappa z)$. The wave falls on the grating from the left only, so $a(0) = a_0$ and $b(L) = 0$:

$$C_1 = a_0, \qquad C_2 = -a_0\tanh\varkappa L, \qquad a(L) = a_0\left(\cosh\varkappa L - \frac{\sinh^2\varkappa L}{\cosh\varkappa L}\right) = \frac{a_0}{\cosh\varkappa L}.$$

Hence $t' = 1/\cosh\varkappa L$ and

$$\boxed{T = \frac{1}{\cosh^2(\varkappa L)}}$$

B.3. In decibels $T_{\min} = -20\log_{10}\cosh\varkappa L$. For $\varkappa L \gg 1$, $\cosh\varkappa L \approx e^{\varkappa L}/2$ and

$$T_{\min} \approx -\frac{20}{\ln 10}\varkappa L + 20\log_{10}2 = -8.69\,\varkappa L + 6.0\ \text{dB}.$$

So the slope of the linear section found in A.8 is $-8.69\varkappa_0$ (the fitted intercept, 5.2 dB, agrees with 6.0 dB):

$$\boxed{\varkappa_0 = \frac{0.743\ \text{dB/mm}}{8.69} = 0.0856\ \text{mm}^{-1} \approx 86\ \text{m}^{-1}}$$

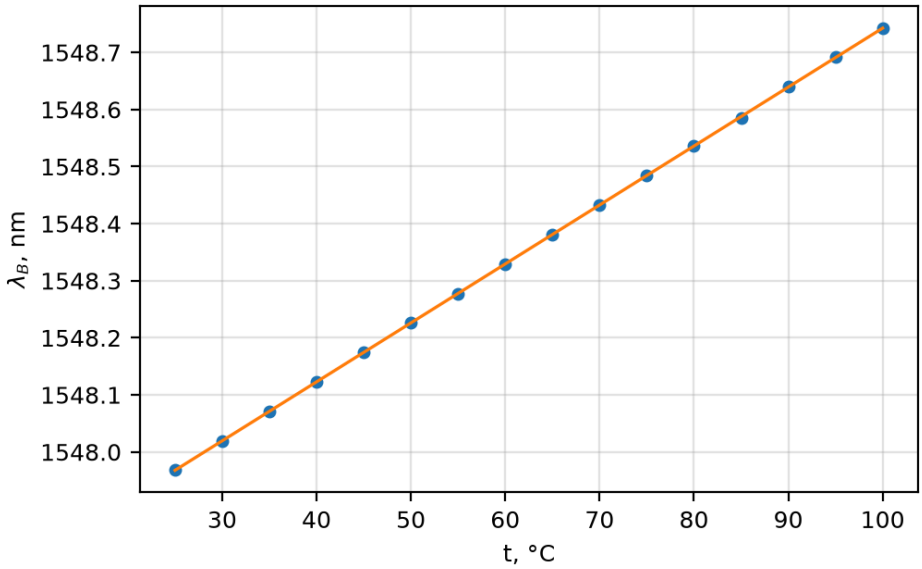
C.1. From $\lambda_B = 2n_c\Lambda$, $\dfrac{\Delta\lambda_B}{\lambda_B} = \dfrac{\Delta\Lambda}{\Lambda} + \dfrac{\Delta n_c}{n_c} = \alpha\Delta t + \dfrac{1}{n_c}\dfrac{dn_c}{dt}\Delta t$, so

$$\boxed{A = \alpha + \frac{1}{n_c}\frac{dn_c}{dt}}$$

C.2. The position of the minimum is read on a plot zoomed to about ± 0.4 nm around the dip.

t, °C	λ_B, nm	t, °C	λ_B, nm
25	1547.9683	65	1548.3809
30	1548.0186	70	1548.4331
35	1548.0717	75	1548.4842
40	1548.1221	80	1548.5357
45	1548.1752	85	1548.5859
50	1548.2264	90	1548.6405
55	1548.2772	95	1548.6922
60	1548.3296	100	1548.7421

C.3.



The slope is $d\lambda_B/dt = 0.01033$ nm/°C and $\lambda_B(25\ ^\circ\mathrm{C}) = 1547.968$ nm, so

$$\boxed{A = \frac{1}{\lambda_B}\frac{d\lambda_B}{dt} = 6.67 \cdot 10^{-6}\ ^\circ\mathrm{C}^{-1}}$$

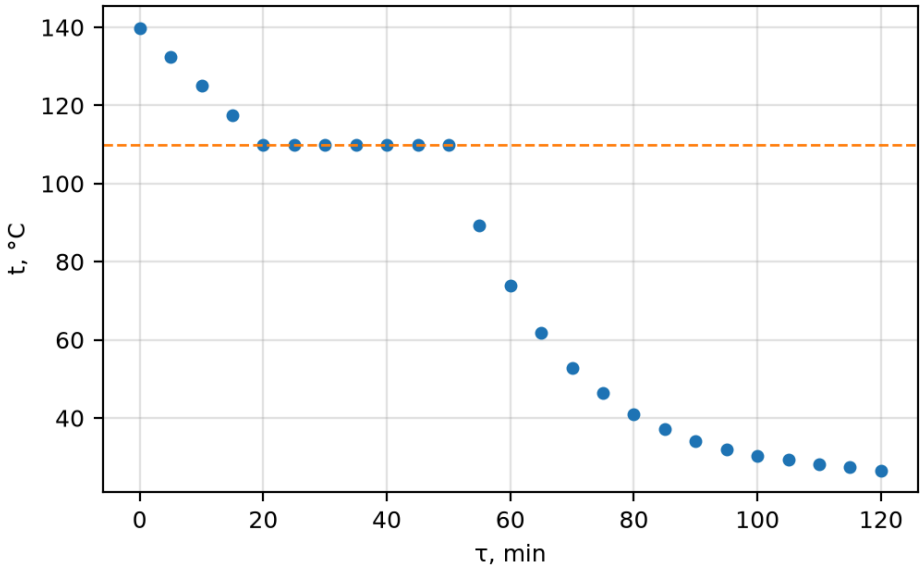
C.4. Repeating the measurement of the same grating at a fixed temperature, the position of the minimum scatters with $\sigma_\lambda \approx 0.0010$ nm (noise of the spectrum). The smallest detectable change is

$$\boxed{\Delta t_{\min} \approx \frac{\sigma_\lambda}{\lambda_B A} \approx 0.1\ ^\circ\mathrm{C}}$$

D.1. The temperature follows from the calibration of part C: $t = 25\ ^\circ\mathrm{C} + \dfrac{\lambda_B - \lambda_B(25\ ^\circ\mathrm{C})}{\lambda_B(25\ ^\circ\mathrm{C})\,A}$.

τ, min	λ_B, nm	t, °C	τ, min	λ_B, nm	t, °C
0	1549.1541	139.8	65	1548.3491	61.9
5	1549.0783	132.5	70	1548.2558	52.9
10	1549.0009	125.0	75	1548.1875	46.3
15	1548.9230	117.5	80	1548.1334	41.0
20	1548.8448	109.9	85	1548.0927	37.1
25	1548.8446	109.9	90	1548.0617	34.1
30	1548.8450	109.9	95	1548.0404	32.0
35	1548.8456	110.0	100	1548.0226	30.3
40	1548.8446	109.9	105	1548.0120	29.3
45	1548.8458	110.0	110	1547.9998	28.1
50	1548.8444	109.8	115	1547.9929	27.4
55	1548.6332	89.4	120	1547.9844	26.6
60	1548.4717	73.8			

D.2.



While the LDPE solidifies, the latent heat keeps the temperature constant: the plateau from $\tau = 20$ to 50 min.

$$\boxed{t_0 = 110\ ^\circ\mathrm{C}}$$

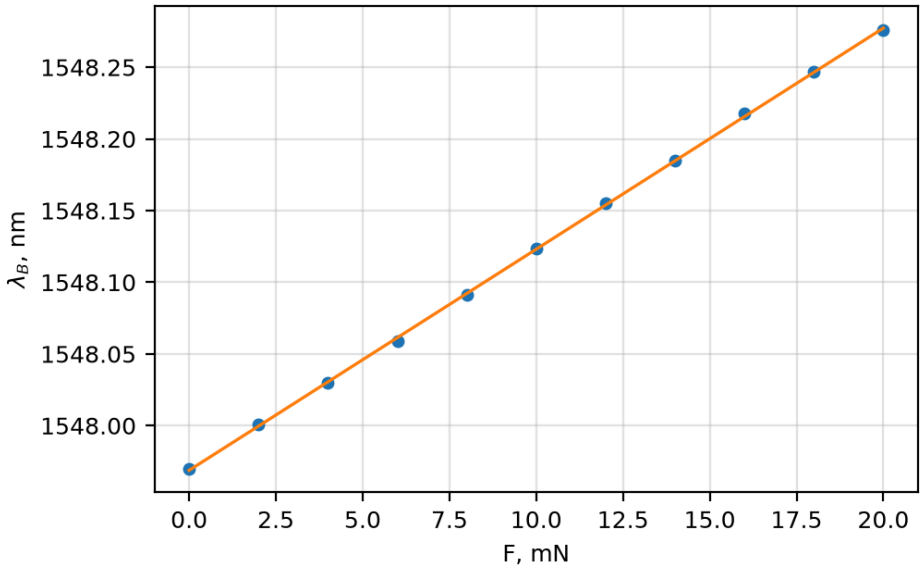
E.1. Under the force the strain is $\varepsilon = \sigma/E = F/(ES)$; the period grows by $\Delta\Lambda/\Lambda = \varepsilon$ and $\Delta n_c = \dfrac{dn_c}{d\varepsilon}\varepsilon$. Then $\dfrac{\Delta\lambda_B}{\lambda_B} = \varepsilon\left(1 + \dfrac{1}{n_c}\dfrac{dn_c}{d\varepsilon}\right)$ and

$$\boxed{B = \frac{1}{ES}\left(1 + \frac{1}{n_c}\frac{dn_c}{d\varepsilon}\right)}$$

E.2.

F, mN	λ_B, nm	F, mN	λ_B, nm
0	1547.9697	12	1548.1549
2	1548.0007	14	1548.1848
4	1548.0298	16	1548.2175
6	1548.0592	18	1548.2470
8	1548.0911	20	1548.2757
10	1548.1233		

E.3.

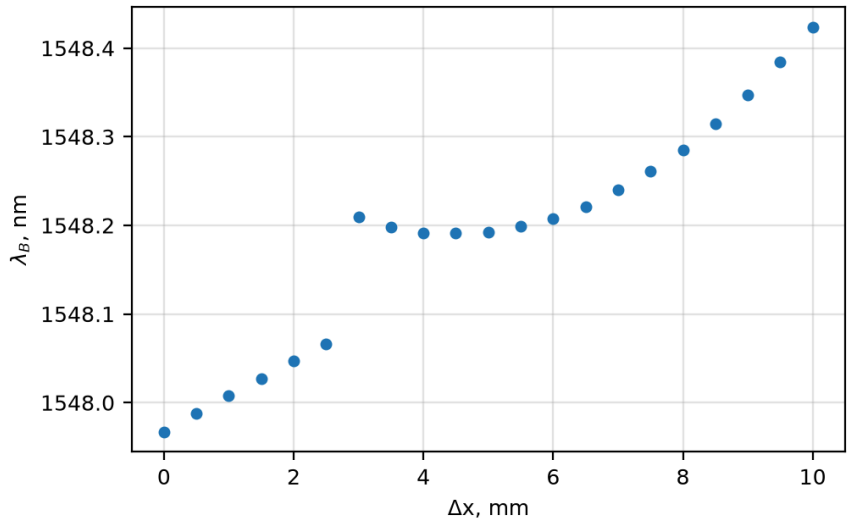


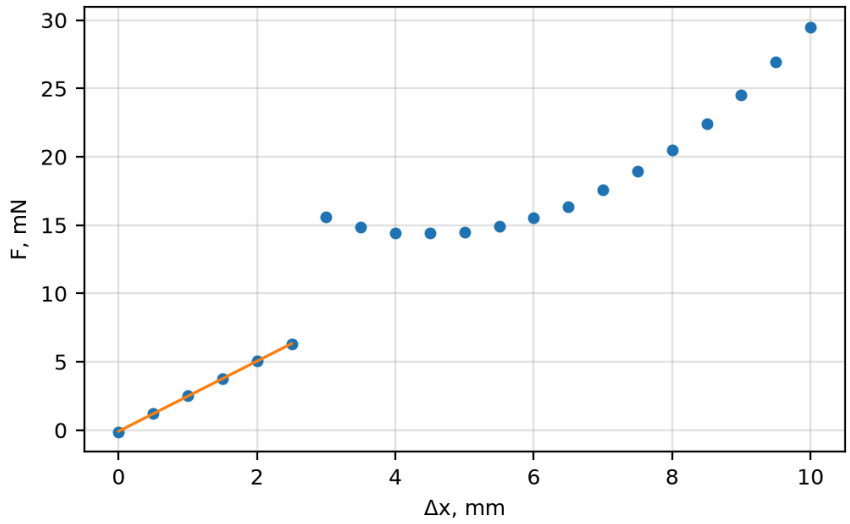
The slope is $d\lambda_B/dF = 0.01542$ nm/mN at $\lambda_B(0) = 1547.969$ nm, so

$$\boxed{B = \frac{1}{\lambda_B}\frac{d\lambda_B}{dF} = 1.00 \cdot 10^{-5}\ \text{mN}^{-1} \approx 1.0 \cdot 10^{-2}\ \text{N}^{-1}}$$

E.4. The force on the fiber is found from the calibration of E.3: $F = \dfrac{\lambda_B - \lambda_B(0)}{\lambda_B(0)\, B}$.

Δx, mm	λ_B, nm	F, mN	Δx, mm	λ_B, nm	F, mN
0.0	1547.9672	−0.11	5.5	1548.1987	14.90
0.5	1547.9876	1.22	6.0	1548.2080	15.51
1.0	1548.0075	2.51	6.5	1548.2211	16.36
1.5	1548.0268	3.76	7.0	1548.2397	17.56
2.0	1548.0471	5.08	7.5	1548.2612	18.96
2.5	1548.0660	6.30	8.0	1548.2853	20.52
3.0	1548.2096	15.61	8.5	1548.3149	22.44
3.5	1548.1982	14.87	9.0	1548.3474	24.54
4.0	1548.1912	14.42	9.5	1548.3848	26.97
4.5	1548.1912	14.42	10.0	1548.4237	29.49
5.0	1548.1920	14.47			



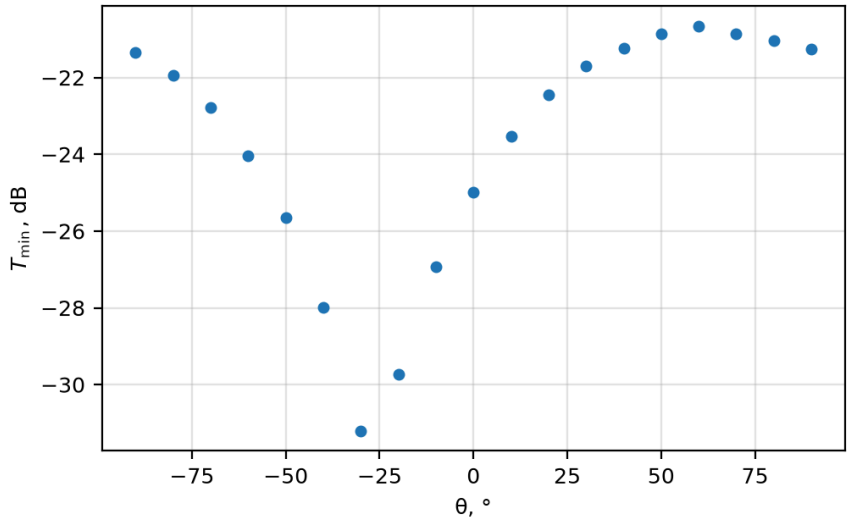


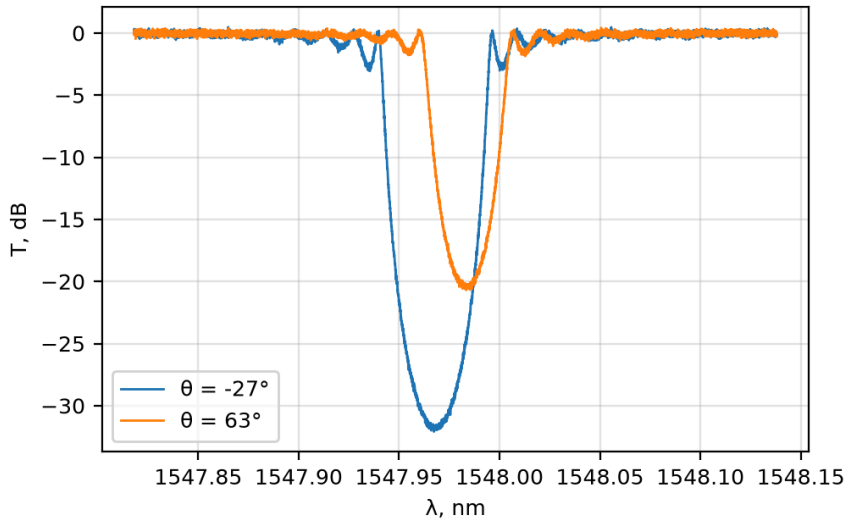
Only the first part of the dependence, $\Delta x \leq 2.5$ mm, is linear; beyond it the force jumps and then grows non-linearly, so the spring no longer obeys Hooke's law there. The slope of the linear part gives

$$\boxed{k = \frac{dF}{d\Delta x} = 2.56\ \text{mN/mm} \approx 2.6\ \text{N/m}}$$

F.1. The light in the grating is the sum of two independent polarizations: the component along the defects with weight $\cos^2(\theta - \theta_0)$ (strength $\varkappa_\parallel$) and the perpendicular one with weight $\sin^2(\theta - \theta_0)$ (strength $\varkappa_\perp$). The depth of the dip is measured as a function of θ:

θ, °	$\lambda_{\min}$, nm	$T_{\min}$, dB	θ, °	$\lambda_{\min}$, nm	$T_{\min}$, dB
-90	1547.9835	−21.35	10	1547.9812	−23.53
-80	1547.9814	−21.95	20	1547.9810	−22.45
-70	1547.9806	−22.79	30	1547.9819	−21.69
-60	1547.9791	−24.03	40	1547.9821	−21.23
-50	1547.9785	−25.64	50	1547.9827	−20.86
-40	1547.9753	−27.99	60	1547.9820	−20.66
-30	1547.9707	−31.20	70	1547.9834	−20.87
-20	1547.9724	−29.74	80	1547.9837	−21.04
-10	1547.9761	−26.92	90	1547.9811	−21.26
0	1547.9793	−24.98			





$T_{\min}(\theta)$ has period 180°. The dip is deepest at $\theta \approx -27^\circ$ and shallowest at $\theta \approx 63^\circ$, where its minimum also moves to a longer wavelength. Since $\varkappa_\parallel < \varkappa_\perp$, the shallow dip is the one for the polarization along the defects:

$$\boxed{\theta_0 \approx 63^\circ \ (\equiv -117^\circ)}$$

F.2. At $\theta = \theta_0$ and $\theta = \theta_0 \pm 90^\circ$ only one grating is seen, and from $T_{\min} = -20 \log_{10} \cosh \varkappa L$ with $L = 50$ mm, $\varkappa = \dfrac{1}{L} \operatorname{arccosh} 10^{-T_{\min}/20}$:

$\theta = 63^\circ$: $T_{\min} = -20.75$ dB, $\lambda_{\min} = 1547.984$ nm; $\theta = -27^\circ$: $T_{\min} = -32.1$ dB, $\lambda_{\min} = 1547.968$ nm.

$$\boxed{\varkappa_\parallel \approx 0.062 \text{ mm}^{-1}, \quad \varkappa_\perp \approx 0.088 \text{ mm}^{-1}, \quad \lambda_{B,\parallel} \approx 1547.984 \text{ nm}, \quad \lambda_{B,\perp} \approx 1547.968 \text{ nm}}$$